REPORT

1.Course Details

Course Name: Artificial Intelligence and Machine Learning
Assignment Title: The Use of GIS in Predicting Soil Nutrition
Student Name: Rithvik Prasad
USN: AMB2215

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2. Problem Statement

Agriculture productivity is highly dependent on soil health, particularly the availability of essential nutrients such as nitrogen (N), phosphorus (P), and potassium (K). Traditional soil testing methods are often time-consuming, expensive, and limited to small geographic areas. This creates a challenge for farmers who need quick and large-scale insights into soil fertility.

The problem addressed in this study is:

"How can Geographic Information Systems (GIS) be used to predict and map soil nutrient levels across large agricultural regions efficiently and accurately?"

This problem is agriculture-oriented as it directly impacts:

- Crop yield optimization
- Fertilizer management
- Sustainable farming practices

For example, instead of manually testing each field, GIS can help predict nutrient levels using spatial data such as location, rainfall, land use, and historical soil records.

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3. Approach / Method

Overview

The approach integrates **GIS with machine learning techniques** to analyze spatial and environmental data for predicting soil nutrient levels.

Methodology Steps

Step 1: Data Collection

- Soil data (N, P, K levels, pH)
- Geographic data (latitude, longitude)
- Environmental factors (rainfall, temperature)
- Satellite imagery (land use patterns)

Step 2: Data pre-processing

- Handle missing values
- Normalize data
- Convert spatial data into usable features

Step 3: GIS Mapping

- Use GIS tools (like QGIS or ArcGIS) to map soil samples
- Generate spatial layers for each variable

Step 4: Model Selection

Common algorithms used:

- Linear Regression
- Random Forest
- Decision Trees
- XGBoost

Justification:
Random Forest is often preferred because:

- It handles non-linear relationships
- Works well with large datasets
- Provides higher accuracy in environmental prediction

Step 5: Model Training

Sample Code Snippet (Python)

```
import pandas as pd

from sklearn.model_selection import train_test_split

from sklearn.ensemble import RandomForestRegressor


# Load dataset

data = pd.read_csv('soil_data.csv')


# Features and target

X = data[['latitude', 'longitude', 'rainfall', 'temperature']]

y = data['nitrogen']


# Split data

X_train, X_test, y_train, y_test = train_test_split(X, y, test_size=0.2)
```

Step 6: Spatial Prediction

- Apply trained model to GIS layers
- Generate prediction maps for nutrient distribution

4. Solution / Output

Expected Results

The system produces:

- Soil nutrient prediction maps
- Region-wise fertility classification
- Recommendations for fertilizer use

Sample Output Table

Location	Nitrogen (Predicted)	Phosphorus	Potassium
Field A	45 kg/ha	Medium	High
Field B	30 kg/ha	Low	Medium
Field C	60 kg/ha	High	High

GIS Output

- Heatmaps showing nutrient distribution
- Color-coded maps:
  - Red → Low fertility
  - Yellow → Medium
  - Green → High

Interpretation

- Farmers can identify nutrient-deficient zones
- Precision agriculture techniques can be applied
- Fertilizer usage becomes optimized and cost-effective

5. Conclusion

The integration of GIS with machine learning provides an efficient and scalable solution for predicting soil nutrition. It reduces dependency on manual soil testing and enables real-time, location-specific decision-making.

Key Takeaways:

- GIS enhances spatial understanding of soil properties
- Machine learning improves prediction accuracy
- Supports sustainable and precision agriculture

Practical Relevance:

- Helps farmers increase crop yield
- Reduces excessive fertilizer usage
- Enables data-driven agricultural planning